

Idukki Monsoon Danger Index

Hyperlocal Monsoon-Risk Intelligence for Inner Idukki, Kerala

Full Project Documentation · Version 1.3

1. Executive summary

The Idukki Monsoon Danger Index turns raw weather data into an operational, plain-language early-warning service for the seven main panchayats of inner Idukki district, Kerala — a region repeatedly hit by landslides and flash floods (2018, 2019, 2020, 2021). The system fetches live conditions and forecasts per locality, computes a transparent four-tier danger index (Low / Moderate / High / Extreme), forecasts the same tiers up to seven days ahead, learns from a 14-year rainfall record to flag heavy-rain danger windows, and pushes the result to the people who need it through a bilingual (English/Malayalam) resident dashboard, colour-coded danger maps, downloadable administrator reports and an SMS alert service that residents subscribe to with their mobile number.

Every layer is honest about its data: modelled estimates are always labelled, live providers (OpenWeatherMap when a key is configured, otherwise Open-Meteo) are named per number, machine-learning results are reported with out-of-sample evaluation, and ward splits are flagged as estimates until a Census table is dropped in. This document covers the full architecture, methodology, evaluation numbers, API surface, the SMS alert flow, and a step-by-step verification of every working flow.

2. Objectives and scope

- Provide a hyperlocal, ward-aware danger level for the 7 main panchayats of inner Idukki (Kumily, Peermedu, Idukki, Adimali, Nedumkandam, Kattappana, Munnar) with their authoritative populations.
- Give non-technical residents plain-language risk levels, the reason for them, and what to do / avoid — in English or Malayalam.
- Forecast danger up to 7 days ahead from the rainfall forecast, using the identical scoring model as the live index.
- Add a machine-learning engine trained on the 2012–2025 local rainfall record, calibrated to catch at least 90% of dangerous heavy-rain windows on data the model has never seen.
- Deliver alerts by SMS with per-user choice of areas, threshold, message language and schedule (instant danger / daily 07:00 / Monday weekly).
- Provide government-grade exports: PDF/DOCX seasonal reports and a REST API.

3. Architecture

Monolith-FastAPI server (api/server.py) that serves both the REST API and the static resident dashboard. Pure-NumPy ML engine, zero-dependency DOCX writer, pure fpdf2 PDF reports. No database: JSON files for caches and subscriptions; /tmp caches for live feeds.

Module	Responsibility
--------	----------------

data/ (fetcher.py, openweather.py, observed.py)	Provider-agnostic live conditions, forecasts, measured history; OpenWeatherMap primary when key set, Open-Meteo otherwise; caching + source tags
index/ (calculator.py, wards.py, map_generator.py)	Danger-index scoring engine, 7-day danger outlook, ward-level micro-zonation, Folium danger map
ml/ (dataset.py, models.py, train.py, predict.py)	LSTM + ridge next-day rainfall and cost-weighted hazard classifier, per locality; out-of-sample evaluation
outlook/ (enso.py, lightning.py)	NOAA CPC ONI ENSO phase + lightning-risk climatology
notify/ (sms.py, store.py, messages.py, scheduler.py)	SMS subscriptions, EN/ML message composer, background alert scheduler, Twilio/Fast2SMS gateway or demo outbox
reporting/report.py	Seasonal risk report PDF (fpdf2) + DOCX (stdlib OOXML)
frontend/static/	Vanilla-JS dashboard: cards, search/filters, 3 trend charts, ward panel, bilingual EN/ML, SMS registration modal
api/server.py	FastAPI application + background threads (ML first-boot training, observed-day closer, SMS scheduler)

4. Data sources

Source	What it supplies	Status
OpenWeatherMap (key in .env)	Current conditions (rain, wind, humidity, cloud, temp) + 5-day/3-h outlook	Primary when OPENWEATHERMAP_API_KEY is set
Open-Meteo (no key needed)	Current conditions, 7-day daily forecast with rain probability, measured past-30-days / history, ERA5 2012–2025 archive for training	Always-available fallback
IMD rain categories	Heavy (64.5 mm) / very heavy (115.5 mm)	Applied, documented daily bands used as hazard thresholds
NOAA CPC ONI	ENSO phase (El Niño / La Niña) for seasonal context	Vendored table, live
KSDMA-style incident register	Historical landslide/flood events near each area (sample of 15 records)	Demo sample — real register is an extension point
LSG Kerala Election-2020	Real 20-ward table for Kumily	Live
Census-2011 populations	Authoritative panchayat totals	Applied per locality

5. Danger-index methodology

The composite score (0–1) combines rainfall-gated weather severity with structural and human factors under a documented NDMA/BIS-style weighting: terrain 40% · soil 25% · exposure (population/wards) 20% · incident history 15%. Rainfall gates the structural risk: without rain the score cannot rise regardless of terrain. The incident-history factor is computed from the register with proximity + severity + recency decay rather than hand-set constants. Tiers map to IMD rain bands and plain-language guidance (what to do / what to avoid), fully translated into Malayalam.

Tier	Meaning	Typical trigger
Low	Normal activity safe; stay alert during monsoon	Calm weather / <35 mm forecast
Moderate	Rain building; slippery roads; avoid heavy-rain travel	35–64 mm expected
High	Landslides/flash floods possible on steep terrain	64–115 mm (IMD “heavy”)

Extreme	Immediate threat to life and property; act now	>115 mm (IMD “very heavy”)
---------	--	----------------------------

6. Hyperlocal ward layer

Panchayat	Population (Census Wards	Ward structure source	
Kumily	33,722	20	LSG Kerala Election-2020 (real)
Peermedu	22,213	19	LSG-typical modelled
Idukki	21,724	19	LSG-typical modelled
Adimali	40,484	20	LSG-typical modelled
Nedumkandam	41,980	21	LSG-typical modelled
Kattappana	42,646	24	LSG-typical modelled
Munnar	32,039	20	LSG-typical modelled

Each ward receives a risk score = panchayat danger raised by that ward’s recorded-incident history, with population apportioned from the authoritative panchayat total. A real Census ward table can be dropped in as `data/static/wards_overrides.csv` (locality, ward_no, ward_name, population, source) — endpoint GET `/wards/{locality}`.

7. Machine-learning engine

Daily rainfall per locality (ERA5 archive via Open-Meteo, 2012–2025) trains three models per panchayat, all pure NumPy: an LSTM forecaster (the proposal’s named architecture), a ridge next-day-rainfall baseline, and a logistic hazard classifier for very-heavy-rain windows (≥ 64.5 mm within 3 days). Misses cost 4x a false alarm in training; each locality’s alert threshold is chosen on the 2018–2019 validation years as the most specific cutoff that still catches $\geq 90\%$ of dangerous windows, then measured honestly on 2020+ data the models never trained on.

Result (test window 2020 → yesterday; full confusion matrices in `ml/models/eval.json`):

Locality	Test AUC	Test recall	Threshold	Heavy-day rate
Kumily	0.938	91.8%	0.025	2.5%
Peermedu	0.943	88.1%	0.038	2.4%
Idukki	0.973	93.7%	0.279	3.2%
Adimali	0.940	92.9%	0.041	4.1%
Kattappana	0.967	93.1%	0.061	3.6%
Munnar	0.941	78.8%	0.129	4.8%
Nedumkandam	0.971	95.5%	0.078	3.7%

Median detection of dangerous windows $\approx 93\%$, AUC 0.94–0.97, validation calibration target met (recall 93% at 14.5% precision). Ridge next-day rainfall: MAE ≈ 7.0 mm vs a daily σ of ≈ 15 mm ($R^2 \approx 0.37$). The live 24-hour outlook shown in every locality panel (“AI outlook”) uses these models and labels its own uncertainty; the operational forecast remains the primary source.

8. Seasonal context and reporting

ENSO phase from the vendored NOAA CPC ONI table (live: Moderate El Niño, ONI +1.39 °C) plus a per-locality lightning-risk climatology give the amber header pill and seasonal advisories. Administrator exports: one PDF and one DOCX per locality or for the whole district (GET `/report`) — headline tier, sub-scores, weather, drivers, incidents and ENSO context, built with `fpdf2` (Unicode font fallback) and a zero-dependency `stdlib` OOXML writer.

9. SMS alert service (resident registration)

The header “SMS Alerts” button (and every locality panel) opens a registration form. A resident enters a validated 10-digit Indian mobile number, chooses the panchayats to watch, the message language(English/Malayalam) and the alert plans:

- Danger alerts — instant SMS when a watched area reaches the chosen threshold (High or Extreme), max once per area per day.
- Daily briefing — every day at 07:00 IST: today’s level and rain plus tomorrow’s forecast for each watched area.
- Weekly outlook — every Monday at 08:00 IST: the week’s worst day per area from the 7-day danger outlook.

A background scheduler (notify/scheduler.py) evaluates subscriptions every 60 s and refreshes the district danger snapshot every 10 min while danger subscriptions exist. Delivery state lives in the subscription record, so restarts never double-send; failed sends retry with backoff. Subscriptions persist in data/subscriptions.json (git-ignored — personal data). Messages are chunked to SMS segment limits and composed per language.

Gateway: demo → live

Without a gateway key the service runs in demo mode: every message is previewed in the UI and recorded to data/notifications/outbox.jsonl so the whole flow is verifiable before connecting a gateway. Add to .env: FAST2SMS_API_KEY (Indian gateway — register a DLT sender/template for transactional SMS) or Twilio keys (TWILIO_ACCOUNT_SID / TWILIO_AUTH_TOKEN / TWILIO_FROM) — real SMS then goes out with zero code changes. GET /notify/status reports which path is live.

Sample messages (demo previews, as generated)

```

△ DANGER ALERT – Kumily, Idukki
Level: High (score 0.72)
Heavy rain on steep terrain
Rain: 78.4 mm · Why: Heavy rain (>64 mm/day) – IMD "heavy rain" category
Stay safe · KSDMA 1077

Idukki · 03 Sep morning briefing
• Kumily: High · 78.4 mm today · tomorrow 45.2 mm
• Munnar: Moderate · 22 mm today
Kerala disaster helpline: 1077

```

10. REST API surface

Method / Path	Purpose
GET /	Dashboard UI (static, bilingual)
GET /health · /summary	Health + district tier/source summary
GET /localities	List with population, ward count, LSG meta
GET /index[/{locality}]	Danger index for all / one panchayat
GET /danger-forecast/{locality}	7-day danger outlook (tier per day, probability, worst day)
GET /trends/{locality}	3 chart series: outlook, 30-day measured, 48-h wind/humidity/temp
GET /wards/{locality}	Ward micro-zones with risk + population
GET /incidents · /map	Historical incidents / interactive Folium map
GET /ml	AI 24-h outlook + eval (AUC, recall, thresholds)
GET /outlook	ENSO phase + lightning advisories

GET /report?locality=&format=pdf|docx

Seasonal report download

GET/POST/DELETE /notify/subscribe · /notify/test · SMS alert registration, testing and audit
/notify/messages · /notify/status

11. Resident dashboard (frontend)

- Professional charcoal theme with restrained forest-green accents (no decorative animation), full English ⇌ Malayalam toggle (English default, remembered per browser; the UI, guidance and SMS messages all translate).
- Search + tier filters and one-click shortcuts: “Most at risk now”, “Worst day this week” (scans all 7 outlooks).
- Per-locality drawer: plain-language guidance, 7-day danger outlook chart (tier-coloured bars), AI model alert, lightning/ENSO, three provider-tagged trend charts, ward-exposure list, nearby incidents, PDF/DOCX downloads, SMS-subscribe shortcut.
- Live indicators: season pill, ● live / ▲ modelled source pill, ENSO pill, provider tags on every number.

12. Verified working flows

Every flow below was executed live on the running system (localhost:8000) and the observed outputs recorded:

- Dashboard render: 7 panchayats with live conditions (Open-Meteo), population and ward counts, district Low headline.
- 7-day danger outlook endpoint returns per-day tier/score/colour/probability + worst_day; a synthetic 45 mm day correctly flips Low → Moderate with landslide drivers.
- ML suite: 7/7 localities trained; /ml predicts next-day rain (~7 mm Kumily); heavy_alert flags respond to thresholds.
- ENSO outlook live: Moderate El Niño · ONI 1.39 °C pill in header; lightning advisories per locality.
- Reports: valid PDF (37.7 KB) and real DOCX downloaded for district and single locality.
- Ward layer: /wards/Kumily returns the 20 real LSG wards with incident-weighted risk; /trends returns all 3 chart series.
- SMS flow: POST /notify/subscribe (EN and ML) → welcome message recorded; GET re-fetches the subscription; scheduler auto-fired a real daily briefing to the test subscription within 60 s; POST /notify/test recorded a preview; DELETE removed it; /notify/messages lists the audit trail.
- Bilingual UI: full Malayalam flip verified across header, hero, cards, filters, drawer, SMS modal — zero console errors.

13. How to run

```
git clone https://github.com/karthikjash/idukki-danger-index
cd idukki-danger-index
.venv/bin/pip install -r requirements.txt # or: pip install -r requirements.txt
.venv/bin/python api/server.py           # dashboard + API on :8000

# optional – better live feed & real SMS:
cp .env.example .env                    # add OPENWEATHERMAP_API_KEY / FAST2SMS_API_KEY (or Twilio)

# regenerate ML models (auto-trained on first boot otherwise):
.venv/bin/python ml/train.py
# regenerate this booklet:
.venv/bin/python docs/generate_project_doc.py
```

Requirements: Python 3.9+, internet for live weather. Docs: README.md ·

14. Limitations and roadmap

- Incident register is a 15-row sample, not the live KSDMA database; terrain is per-locality constant, not SRTM elevation/slope.
- Open-Meteo/ERA5 weather rather than IMD gridded data (IMD requires registration); no validation against actual 2018–2020 outcomes yet (a backtest script is the next step).
- Ward structures for six panchayats are LSG-typical models until Census ward tables are dropped in via wards_overrides.csv.
- SMS alerting is a prototype awaiting a gateway key for real delivery; danger alerts currently trigger on the computed tier, refreshed every 10 minutes.

15. Document control

Item	Value
------	-------

Project	Idukki Monsoon Danger Index (github.com/karthikjash/idukki-danger-index)
Document version	1.3 — generated September 2026
Reproducibility	Run docs/generate_project_doc.py; ML numbers read live from ml/models/eval.json
Coverage tracking	docs/PROPOSAL_COVERAGE.md maps every proposal item to status